

Periodic viscous Burgers as a positive projective heat flow

Global phase portrait, first-mode asymptotics, and a Route-A boundary

HCS-C195 research artifact

27 August 2026 · revision 2

Abstract

For every viscosity $\nu > 0$, circumference $L > 0$, mean $m \in \mathbb{R}$, and $s > 3/2$, we identify the periodic viscous-Burgers semiflow on its fixed-mean H^s leaf with the strictly positive projectivization of a drift-heat semigroup. This gives a global physical-time solution and instant smoothing. Every orbit converges to the unique constant equilibrium, so no nonconstant periodic or recurrent point exists. The first active Fourier mode of the positive lift gives an exact leading expansion, and the entire linearized spectrum is explicit. A 24-case exact oracle checks algebraic regressions but is not used as proof. The transformation is classical Hopf–Cole mathematics; the resulting linear lift is only a formal Route-A hint and supplies no arithmetic structure.

1 Frozen phase leaf and the correct linear state

On $\mathbb{T}_L = \mathbb{R}/L\mathbb{Z}$, consider

$$u_t + uu_x = \nu u_{xx}, \quad X_m^s = \{u \in H^s(\mathbb{T}_L; \mathbb{R}) : L^{-1} \int_{\mathbb{T}_L} u \, dx = m\}. \quad (1)$$

The clock is the source physical time. Define

$$\mathbb{P}_+^{s+1} = \{w \in H^{s+1}(\mathbb{T}_L; \mathbb{R}) : \min w > 0\} / \mathbb{R}_{>0}, \quad \Phi_m([w]) = m - 2\nu \partial_x \log w. \quad (2)$$

The quotient is essential: positive scalar multiples give the same Burgers state. If $V_x = u - m$ and V is the unique zero-mean periodic primitive, then

$$\Phi_m^{-1}(u) = [e^{-V/(2\nu)}]. \quad (3)$$

Sobolev composition makes (2)–(3) smooth inverse maps for $s > 3/2$.

In fixed x , the autonomous linear semigroup is

$$K_t = e^{t(\nu \partial_x^2 - m \partial_x)}, \quad K_t w(x) = (e^{\nu t \partial_x^2} w)(x - mt). \quad (4)$$

Thus pure heat appears after the Galilean coordinate $y = x - mt$; it is not mislabelled as the autonomous fixed-coordinate flow.

Theorem 1 (global projective Cole–Hopf conjugacy). *For all parameters above,*

$$S_t \Phi_m([w]) = \Phi_m([K_t w]), \quad t \geq 0. \quad (5)$$

The cone is invariant, every Burgers orbit is global, and $S_t u$ is smooth for every $t > 0$. Moreover $S_t u \rightarrow m$ in H^s . Hence $u \equiv m$ is the unique equilibrium, periodic point, and recurrent point on X_m^s , where recurrence means $S_{t_j} u \rightarrow u$ for some $t_j \rightarrow \infty$.

Write $\kappa_k = 2\pi k/L$ and $w_0 = \sum a_k e^{i\kappa_k x}$. If $u_0 \neq m$, let r be the first active absolute mode and r_2 the next one (or ∞). Then, with $\eta = \min(2\nu\kappa_r^2, \nu\kappa_{r_2}^2) > \nu\kappa_r^2$,

$$S_t u_0 - m = -\frac{2\nu}{a_0} \sum_{|k|=r} i\kappa_k a_k e^{-\nu\kappa_k^2 t} e^{i\kappa_k(x-mt)} + O_{H^s}(e^{-\eta t}). \quad (6)$$

Consequently the exact logarithmic decay exponent is $\nu\kappa_r^2$. The complexified linearization at m , with domain $H^{s+2} \subset H^s$, has complete spectrum

$$\lambda_k = -\nu\kappa_k^2 - im\kappa_k, \quad k \in \mathbb{Z}, \quad (7)$$

with $k = 0$ removed on the fixed-mean tangent space.

2 Proof mechanism

If $w_t = \nu w_{xx} - mw_x$ and $u = m - 2\nu w_x/w$, direct differentiation gives $u_t + uu_x - \nu u_{xx} = 0$; conversely (3) recovers the same projective linear orbit. The periodic heat kernel preserves strict positivity, proving (5) and global smoothing.

Fourier diagonalization gives

$$K_t w_0 = a_0 + \sum_{k \neq 0} a_k e^{-\nu\kappa_k^2 t} e^{i\kappa_k(x-mt)}, \quad a_0 > 0. \quad (8)$$

Hence $K_t w_0 \rightarrow a_0$ in H^{s+1} , so $S_t u_0 \rightarrow m$ in H^s . A returning subsequence must have both limits u and m , proving the recurrence claim. Separating the $|k| = r$ pair in (8), the next linear tail decays at $\nu\kappa_{r_2}^2$, while the first nonlinear term in $(a_0 + h)^{-1}$ decays at $2\nu\kappa_r^2$. This proves (6); its leading pair is nonzero and translation preserves its H^s norm. Finally $h_t = \nu h_{xx} - mh_x$ diagonalizes in Fourier modes. Compact resolvent gives the completeness asserted in (7).

3 Exact oracle, ownership, and stopping boundary

At the regression normalization $L = 2\pi$, 24 strictly positive rational trigonometric lifts certify 24 zero cleared-denominator PDE residuals, 60 reality cells, 24 exact semigroup compositions, 24 leading-mode rows, and 408 spectral cells. A producer-independent checker executes 1,490 assertions; a separate SymPy program executes 129 checks on nine cases; byte replay passes; and 22 repaired-hash plus one stale-hash mutations are rejected. These finite rows are regression sentinels only—the all-function theorem is proved above.

certificate layer	exact count	logical role
positive Laurent lifts	24	sign, gap, and parameter regression
generator residuals	24 zero	cleared- w^3 PDE identity
semigroup compositions	24 zero	physical-time composition sentinel
linear spectrum	408 cells	modes -8 through 8 , not completeness proof
checker / SymPy	1,490 / 129	separate implementations and selected expansions
semantic mutations	22 + 1 rejected	repaired-hash semantics plus stale hash

The substitution and viscous-Burgers solution method are classical work of Hopf [1] and Cole [2]; no discovery or priority is claimed here. The source family has no intrinsic rational-prime carrier, no nonconstant recurrent or periodic orbit, no primitive divisor, and no target analytic structure. The exact verdict is

$$(A0, A1, A2, A3, A4) = (A0_FAIL, A1_FAIL, A2_FAIL, \\ A3_FAIL, A4_FORMAL_HINT).$$

The scope literal is `NO_BAD_EULER_OR_ROOT_NUMBER`. Route A is rejected and Route B is not authorized. No Euler product or factor, root number, automorphy statement, target functional equation, or Hilbert–Pólya operator is asserted.

Release boundary. The theorem covers all $\nu > 0, L > 0, m \in \mathbb{R}$ and $u_0 \in X_m^s$, $s > 3/2$, but not the inviscid limit, nonperiodic domains, or rougher phase spaces. The physical clock is never reparametrized. The exact oracle uses no target table; its rational heat/translation pairs probe a universal two-parameter multiplier and are not labelled as points on the physical one-parameter curve. Those normalized finite Fourier rows cannot establish the infinite-dimensional result. The positive projective heat generator remains a source operator, not a quantization claim.

References

- [1] E. Hopf, *The partial differential equation $u_t + uu_x = \mu u_{xx}$* , Commun. Pure Appl. Math. 3 (1950), 201–230. doi:10.1002/cpa.3160030302.
- [2] J. D. Cole, *On a quasi-linear parabolic equation occurring in aerodynamics*, Quart. Appl. Math. 9 (1951), 225–236. JSTOR 43633894.